

Exp :27 Binary Semaphore in SystemC

Program:

```
// Code your testbench here.

// Uncomment the next line for SystemC modules.

// #include <systemc>

#include <systemc.h>

SC_MODULE(binary_semaphore) {

    int sem;

    sc_event ev;

    void task1() {

        while(true) {

            wait(1, SC_SEC);

            if(sem == 1) {

                sem = 0;

                cout << "Task1 Entered Critical Section at " << sc_time_stamp() << endl;

                wait(2, SC_SEC);

                sem = 1;

                ev.notify();

            }

            else {

                wait(ev);

            }

        }

    }

    void task2() {

        while(true) {

            wait(2, SC_SEC);
```

```

    if(sem == 1) {
        sem = 0;

        cout << "Task2 Entered Critical Section at " << sc_time_stamp() << endl;

        wait(1, SC_SEC);

        sem = 1;

        ev.notify();
    }
    else {
        wait(ev);
    }
}
}

```

```

SC_CTOR(binary_semaphore) {
    sem = 1;

    SC_THREAD(task1);

    SC_THREAD(task2);
}

};

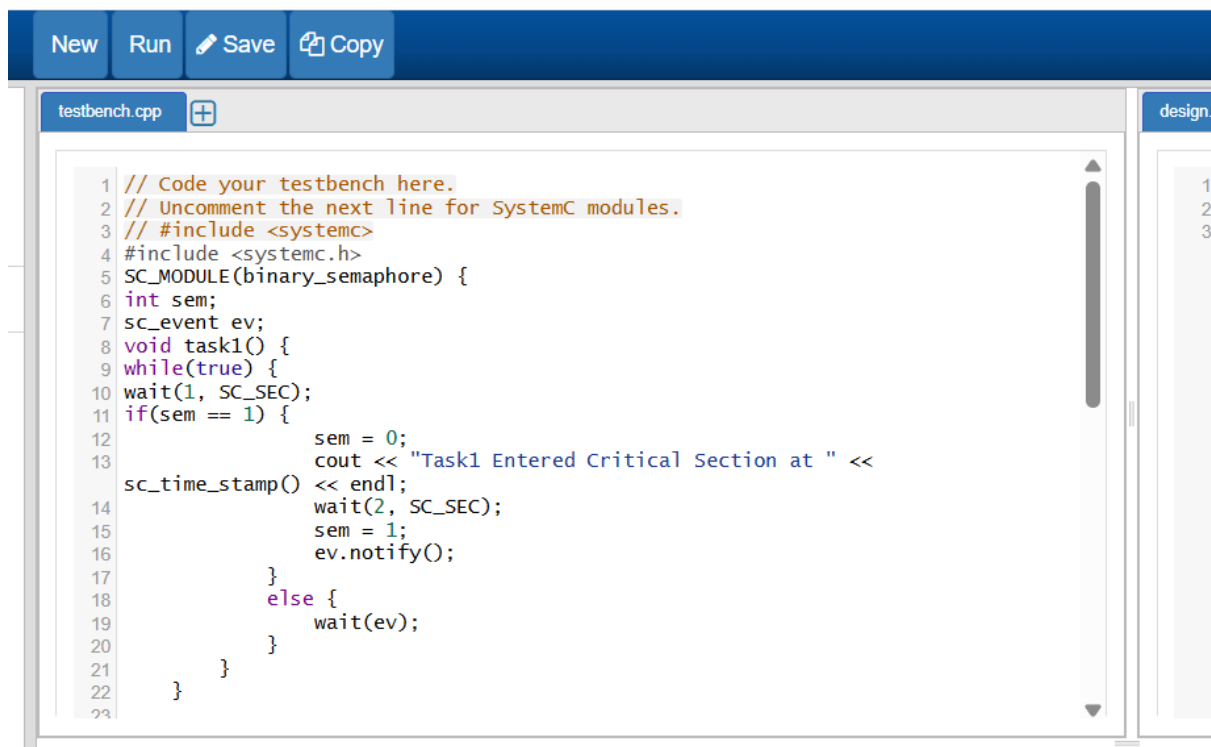
int sc_main(int argc, char* argv[]) {
    binary_semaphore obj("Binary_Semaphore");

    sc_start(10, SC_SEC);

    return 0;
}

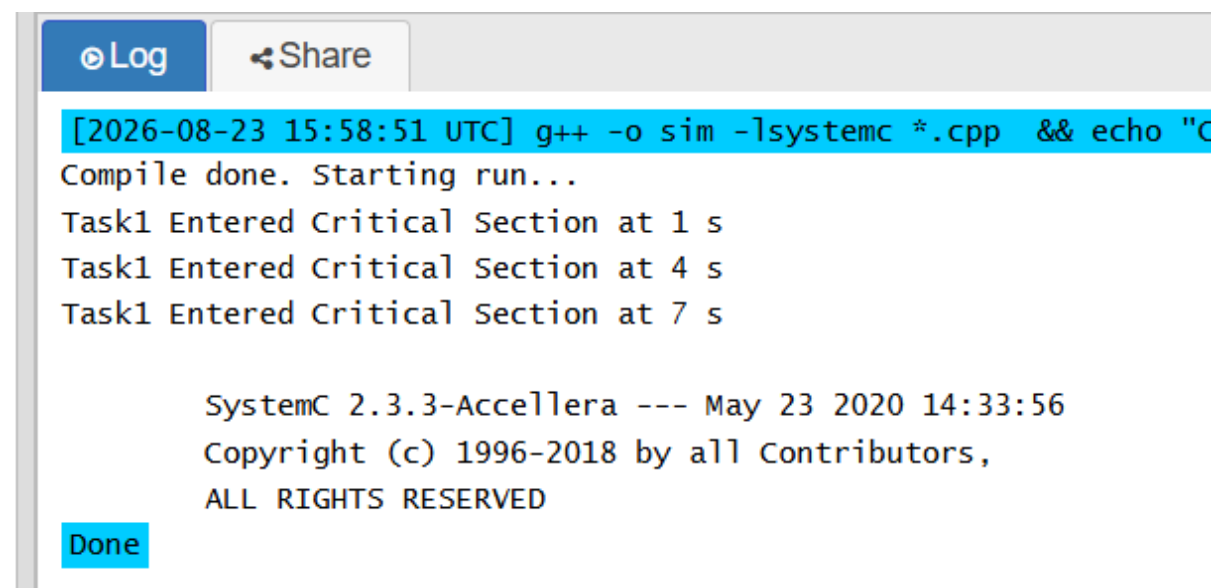
```

}



```
1 // Code your testbench here.
2 // Uncomment the next line for SystemC modules.
3 // #include <systemc>
4 #include <systemc.h>
5 SC_MODULE(binary_semaphore) {
6     int sem;
7     sc_event ev;
8     void task1() {
9         while(true) {
10             wait(1, SC_SEC);
11             if(sem == 1) {
12                 sem = 0;
13                 cout << "Task1 Entered Critical Section at " <<
14                 sc_time_stamp() << endl;
15                 wait(2, SC_SEC);
16                 sem = 1;
17                 ev.notify();
18             }
19             else {
20                 wait(ev);
21             }
22         }
23     }
```

OutPut



```
[2026-08-23 15:58:51 UTC] g++ -o sim -lsystemc *.cpp && echo "C
Compile done. Starting run...
Task1 Entered Critical Section at 1 s
Task1 Entered Critical Section at 4 s
Task1 Entered Critical Section at 7 s

SystemC 2.3.3-Accellera --- May 23 2020 14:33:56
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```

Done